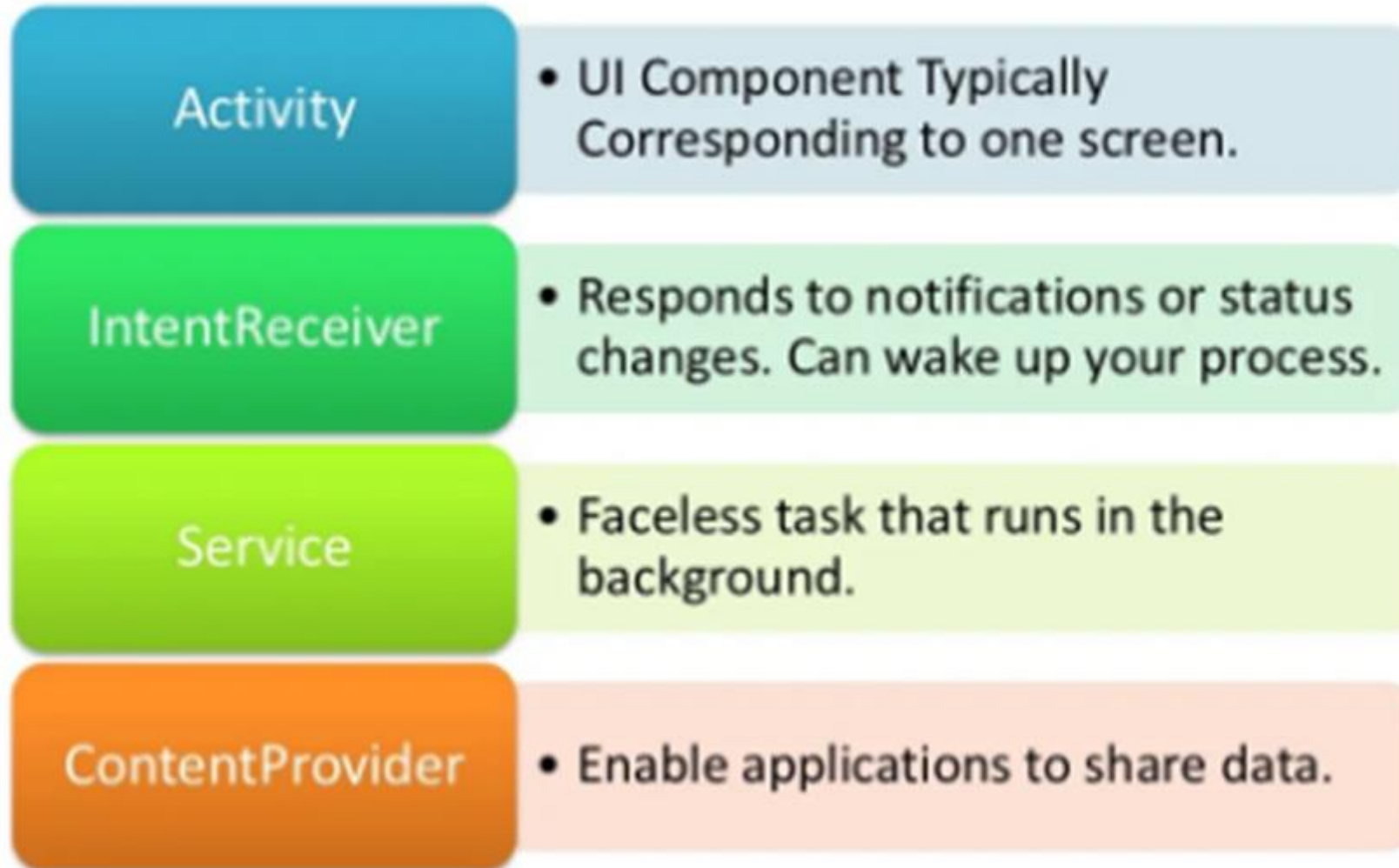


Android Application Building Blocks



Android Application Building Blocks



Android Application Anatomy

Activities

1. Provides **User Interface**
2. Usually represents a **Single Screen**
3. Can contain one/more **Views**
4. **Extends** the **Activity** Base class

Services

1. **No User Interface**
2. Runs in **Background**
3. **Extends** the **Service** Base Class

Application= Set of Android Components

Intent/Broadcast Receiver

1. Receives and Reacts to broadcast **Intents**
2. No UI but **can start** an Activity
3. **Extends** the **BroadcastReceiver** Base Class

Content Provider

1. Makes application data available to other apps
2. Data stored in SQLite database
3. **Extends** the **ContentProvider** Base class

Android Application Core Building Blocks

- An **activity** is a class that represents a single screen. It is like a Frame in AWT.
- A **view** is the UI element such as button, label, text field etc. Anything that you see is a view.
- **Intent** is used to invoke components. It is mainly used to:
 - Start the service
 - Launch an activity
 - Display a web page
- **Service** is a background process that can run for a long time.
- There are two types of **services** local and remote. Local service is accessed from within the application whereas remote service is accessed remotely from other applications running on the same device.
- **Content Providers** are used to share data between the applications.
- **Fragments** are like parts of activity. An activity can display one or more fragments on the screen at the same time.
- **AndroidManifest.xml**: It contains information's about activities, content providers, permissions etc. It is like the web.xml file in Java EE.
- **Android Virtual Device (AVD)**: It is used to test the android application without the need for mobile or tablet etc. It can be created in different configurations to emulate different types of real devices.



Android Application Core Building Blocks

Broadcast Receivers

1. A *broadcast receiver* is a component that responds to system-wide Broadcast announcements.
2. Many broadcasts originate from the system—for example, a Broadcast announcing that the *screen has turned off, the battery is low, or a picture was captured or an SMS is received*.
3. Applications can also initiate broadcasts—for example, to let other Applications know that some data has been downloaded to the device and is available for them to use.
4. Although broadcast receivers don't display a user interface, they may **create a status bar notification** to alert the user when a broadcast event occurs.
5. More commonly, though, a broadcast receiver is just a "gateway" to other components and is intended to do a very minimal amount of work. For instance, it might initiate a service/or start an activity to perform some work based on the event.

Android Application

Component Types

- **Activity** is a single screen with a user interface.
- **Service** performs long-running tasks in background.
- **Content Provider** manages shared set of data.
- **Broadcast Receiver** responds to system-wide announcements.
 - Broadcast in android is **the system-wide events that can occur when the device starts, when a message is received on the device or when incoming calls are received, or when a device goes to airplane mode, etc.** Broadcast Receivers are used to respond to these system-wide events.

Android Application

Component Types

- For each app component that you create in your app, you must declare a corresponding XML element in the manifest file:
 - **<activity>** for each subclass of Activity.
 - **<service>** for each subclass of Service.
 - **<receiver>** for each subclass of BroadcastReceiver.
 - **<provider>** for each subclass of ContentProvider.

Android Application Like As ...

Android OS as Hotel:

- Your **App** is the guest.
- The **Android System** is the hotel manager / Administration / Staff.
- Services are available when you request them (**Intents**).
- In the foreground (**Activities**) such as registration.
- In the background (**Services**) such as laundry.
- Calls you when a package (Parcel) has arrived (**Broadcast Receiver**).
- Access the city's tour companies (**Content Provider**).